Assignment #5 for Numerical Analysis

Deadline: Mar. 31

1. (30 points) Consider the system

$$\begin{bmatrix} 1 & 1 \\ 1+\delta & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2+\delta \end{bmatrix}$$

   (a) Find the condition number of the coefficient matrix as a function of $\delta > 0$.

   (b) Find the error magnification factor for the approximate root $x_a = [-1, 3+\delta]$.
   ($cond(A) = ||A|| * ||A^{-1}||, ||A||_\infty$=maximum absolute row sum)
   (error magnification factor=$\frac{relative\ forward\ error}{relative\ backward\ error}$)

2. (30 points) Rearrange the equations to form a strictly diagonally dominant system. Apply two steps of the Jacobi and Gauss–Seidel Methods from starting vector $[0, \cdots, 0]$.

$$\begin{aligned} u + 4v &= 5 \\ v + 2w &= 2 \\ 4u + 3w &= 0 \end{aligned}$$

3. (40 points) Apply Gauss–Seidel Method to the system $\boldsymbol{Ax} = \boldsymbol{b}$. Denote $x_i^{(k+1)}$ as the i-th entry of $\boldsymbol{x}^{(k+1)}$, and $\boldsymbol{\varepsilon}^{(k)} = \boldsymbol{x}^{(k)} - \boldsymbol{x}^*$, where $\boldsymbol{x}^*$ is the exact solution.

   (a) Prove that $x_i^{(k+1)} = x_i^{(k)} + \frac{r_i^{(k+1)}}{a_{ii}}$, where

$$r_i^{(k+1)} = b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(k+1)} - \sum_{j=i}^{n} a_{ij} x_j^{(k)}.$$

   (b) Prove that $\varepsilon_i^{(k+1)} = \varepsilon_i^{(k)} - \frac{R_i^{(k+1)}}{a_{ii}}$, where

$$R_i^{(k+1)} = \sum_{j=1}^{i-1} a_{ij} \varepsilon_j^{(k+1)} + \sum_{j=i}^{n} a_{ij} \varepsilon_j^{(k)}.$$